Day 1 - Foundations & Linear Networks

What deep learning is, where it comes from, and the simplest network that learns

Annika Stiehl · Deep Learning · M.Sc. · SRH

Today

Time	
09:00 - 10:30	Welcome, getting to know each other, organisation, exam • What is deep learning? History & motivation
10:45 - 12:15	Perceptron & representational power • Linear regression as a neural network, gradient descent
13:00 - 14:30	Lab 1: Colab & PyTorch, linear regression from scratch → autograd → `nn.Module`
14:45 - 16:00	Lab 1b: EEG data pipeline (CHB-MIT, SeizeIT2) • Exam overview • Q&A

Welcome & Organisation

About me

- Annika Stiehl
- PhD research: deep learning for EEG seizure detection
- Interests: biomedical signal processing, robust evaluation, wearables
- Hochschule Ansbach (Ansbach University of Applied Sciences)
- ✉️ annika.stiehl@hs-ansbach.de

🧠 Running theme

Many examples in this course come from my field: detecting epileptic seizures in EEG. You are free to work on images or other data in your project.

Getting to know each other

menti.com/al53jyhjr96r

👥 Join the Mentimeter (5 min)

1. scan the code or open the link on your phone
2. answer the questions
3. we look at the results together



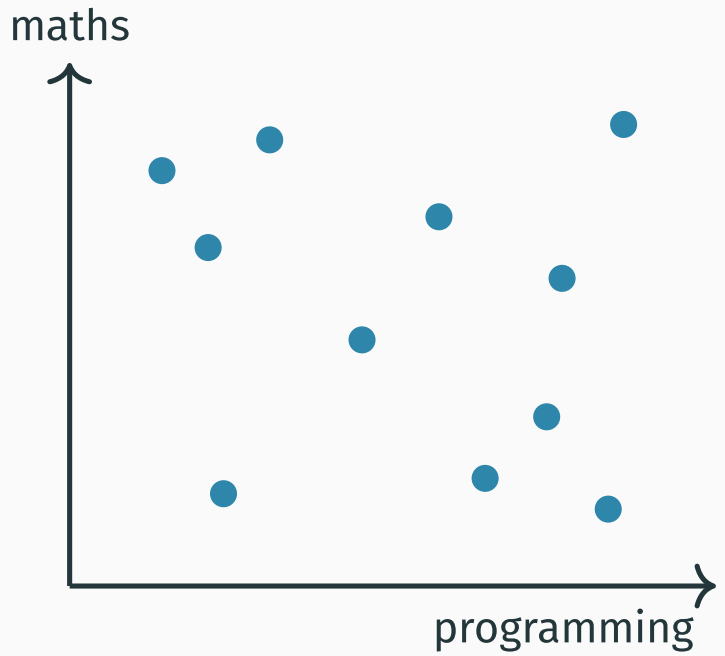
Activity: you are the training data

Stand up and place yourself in the room:

- **left → right**: programming experience (never → every day)
- **front → back**: comfort with maths (derivatives, matrices, probability)

Look around: clusters? outliers? an empty region?

Then one after the other: your **name** and your study programme - one sentence each.



👥 Your turn

You just formed a 2D dataset. How would you choose teams of two that cover the whole space?

What you will be able to do after this course

- **Understand** how neural networks learn: loss, gradients, backpropagation, optimisation
- **Implement** MLPs, CNNs, RNNs/LSTMs, autoencoders, VAEs, GANs and GNNs in PyTorch
- **Train** them well: regularisation, normalisation, learning-rate schedules, transfer learning
- **Evaluate** them honestly: proper splits, suitable metrics, statistical tests
- **Build on** a current research paper: baseline, the paper's idea, your own idea - and present it as paper + poster

🔑 Key takeaway

Priority in this course: understanding and coding. Every concept is derived on the slides and then implemented by you in the lab.

Course roadmap

Day	Topic	Lab	Project
1	Foundations & linear networks	PyTorch, linear regression, first EEG	kick-off
2	MLPs & backpropagation	backprop from scratch, MLP on EEG features	groups + 3 papers
3	Training deep networks & evaluation	optimisers, regularisation, leakage, statistics	proposal
4	Convolutional neural networks	CNN on CIFAR-10, 1D CNN on CHB-MIT	data pipeline
5	Transfer learning & domain shift	fine-tuning on MedMNIST	check-in: A running
6	RNN, LSTM, GRU	long memory, CNN+GRU on SeizeIT2	A results, B running
7	Autoencoders & generative models	AE, VAE, EEG anomaly detection	B results, C running, poster draft
8	Graph neural networks	GCN on Cora and EEG graphs	poster session

A typical day

- **Morning:** two lecture blocks - intuition / math / code snippets
- **Afternoon:** lab in Google Colab - notebook with exercises (`# TODO`)
- **From day 4:** last block partly project work

💻 Notebooks

- `dayN_*.ipynb` : with gaps
- solutions are released after the lab
- open in Colab, select a T4 GPU

Slides & notebooks



github.com/annikastiehl/srh-deep-learning

👥 Do it now (5 min)

1. scan the code, open the repository
2. click **Open in Colab** for today's notebook
3. File → Save a copy in Drive
4. Runtime → Change runtime type → T4 GPU
5. run the first cell - does it print `device: cuda`?

Assessment: practical work in teams of two (100 %)

A challenge - the same image dataset for every team:

1. run and understand a **baseline** CNN
2. **improve** it with at least two well-motivated ideas
3. **evaluate** properly (metrics, several seeds, confidence intervals)
4. make it usable in a small **web app**
5. write a short **report** (IEEE format) and present a **poster** on **11 November**

All details: **day 4** (challenge kick-off) and the challenge handout.

AI tools

AI assistants are allowed - but you must disclose them (AI use statement) and be able to explain every line.

Tools

Google Colab

- free GPU (usually an NVIDIA T4, 16 GB)
- sessions time out - save to Google Drive
- limited GPU hours per day: debug on CPU / small data first

PyTorch

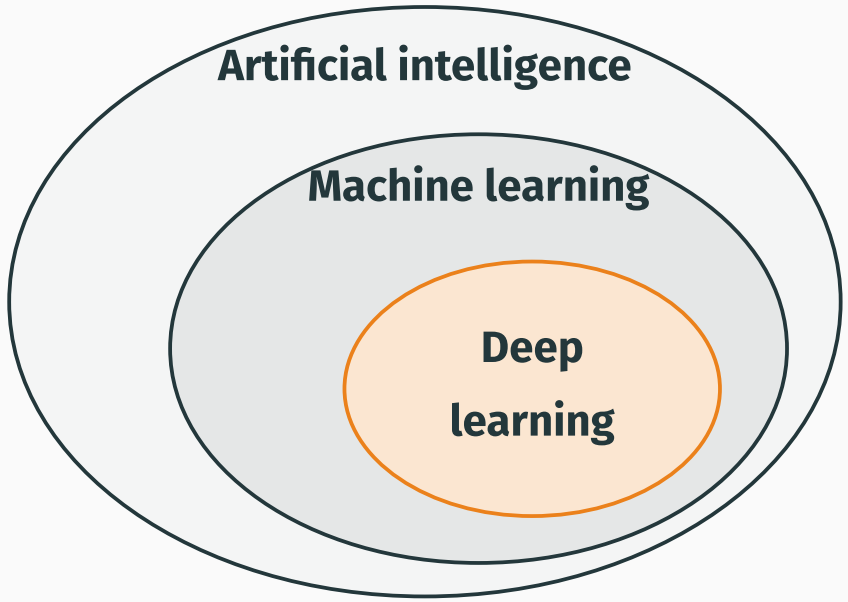
- tensors + automatic differentiation + GPU
- the de-facto standard in research code
- most research papers release PyTorch code

📚 Literature (all free online except Goodfellow print)

Zhang et al.: Dive into Deep Learning (d2l.ai) • Goodfellow, Bengio & Courville: Deep Learning (2016) • Nielsen: Neural Networks and Deep Learning • Prince: Understanding Deep Learning (2023) • Stanford CS231n notes

What is Deep Learning?

AI, machine learning, deep learning



- **AI**: systems that perform tasks requiring "intelligence" (also rule-based)
- **ML**: learn the mapping from data instead of programming it
- **DL**: ML with deep neural networks - many layers of learned, differentiable transformations

Your turn: AI, ML or DL?

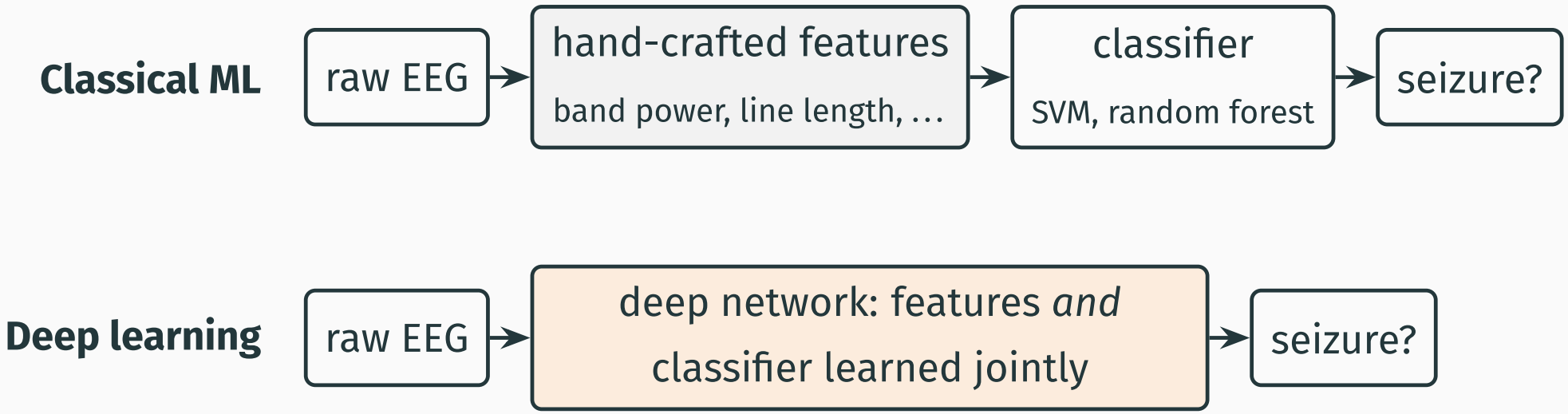
In pairs (3 min): where in the diagram does each system belong - and why?

1. Deep Blue beats Kasparov at chess (1997)
2. AlphaZero learns chess by self-play (2017)
3. route planning in a navigation app
4. seizure detection: band-power features + SVM
5. face unlock on your phone
6. a chatbot like ChatGPT or Claude

👥 Your turn

Now add your own examples: which AI systems did you use this week? Where do they belong?

The key difference: learned representations



- Classical: expert knowledge goes into the features
- Deep learning: expert knowledge goes into the architecture, data and loss; features are learned
- Price: more data, more compute, less transparency

Try it: train a network in your browser

Teachable Machine - teachablemachine.withgoogle.com

1. Get started → Image project → Standard
2. 2-3 classes, e.g. thumbs up / thumbs down / nothing; record about 50 webcam images each
3. Train model, then test it live
4. now break it: other background, other light, your neighbour's hand

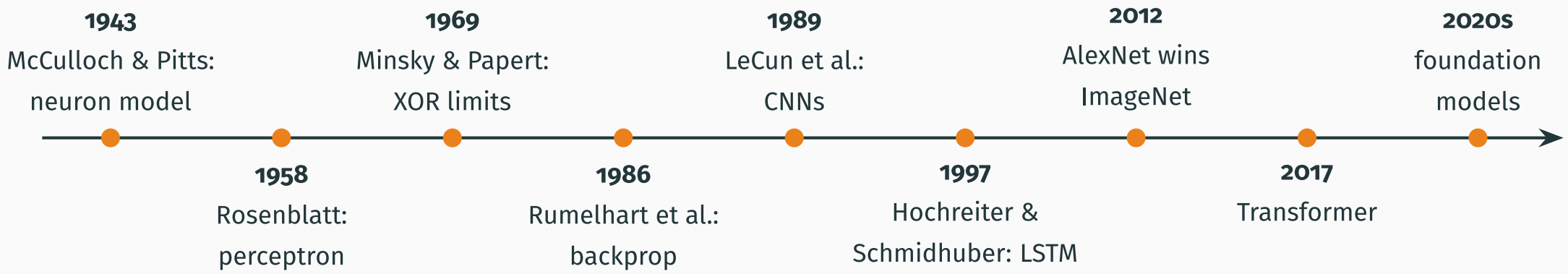
👥 Your turn (20 min)

Runs entirely in your browser - no GPU, no install. No webcam? Pair up, or use an Audio project.

❓ Think about it

Which features did the network learn? Why does it fail when the background changes?

A short history of neural networks



- Two "AI winters" (1970s, late 1980s-90s) - the ideas are old, what changed after 2010 was data, compute and engineering
- 2024: Nobel Prize in Physics to Hopfield and Hinton for foundational work on neural networks

Why now? Three ingredients

🗄️ Data

ImageNet (1.2 M labelled images), web-scale text, large clinical archives (e.g. TUH EEG corpus)

🖥️ Compute

GPUs: thousands of parallel multiply-adds. Neural networks are mostly matrix products.

🛠️ Algorithms & software

ReLU, dropout, BatchNorm, Adam, residual connections, autograd frameworks (PyTorch, TensorFlow, JAX)

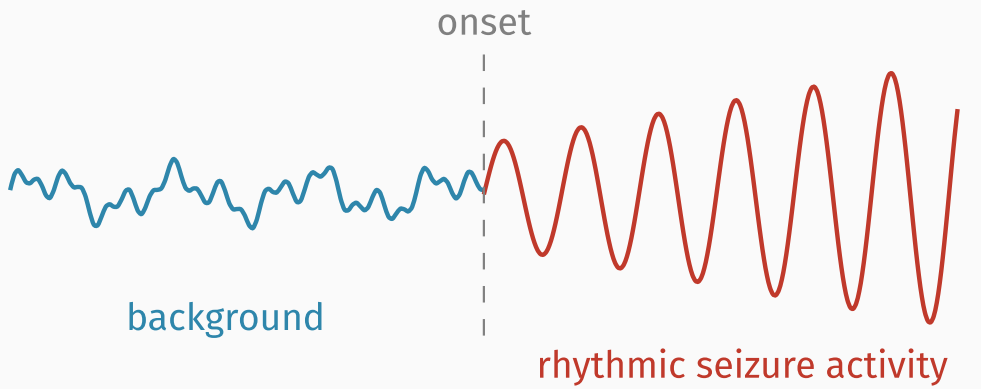
❓ Think about it

AlexNet (2012) reduced the ImageNet top-5 error from 26.2 % to 15.3 %. Which of the three ingredients made that possible?

Krizhevsky, Sutskever & Hinton, NeurIPS 2012

Running example: epileptic seizure detection in EEG

- Epilepsy affects ≈ 50 million people worldwide
- Diagnosis: days of video-EEG monitoring
- Neurologists review hours of EEG by eye
- Goal: detect seizures automatically - in hospital and with wearables at home



🧠 Why this is a great deep learning problem

Long multichannel time series • extreme class imbalance (seizures are < 1 % of the time) • large variability between patients • artifacts • noisy expert labels ⇒ every topic of this course shows up.

Datasets in this course

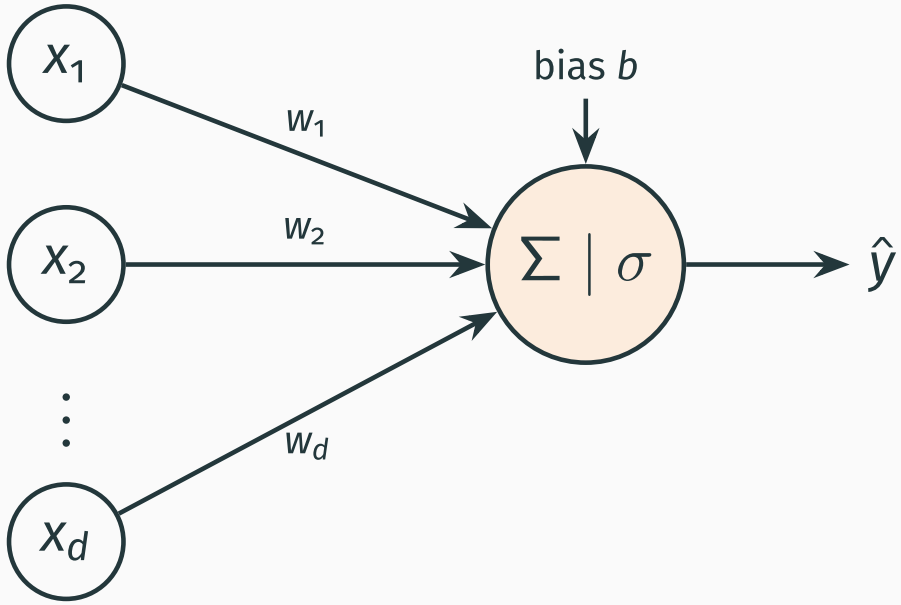
Dataset	Content	Used on
CHB-MIT	scalp EEG of 22 paediatric patients (23 cases), 23 bipolar channels, 256 Hz, 198 seizures	days 1-4, 7, 8
SeizeIT2	wearable behind-the-ear EEG (2 channels) + ECG, EMG, movement; 125 patients, 5 European centres, 886 seizures	day 6
MNIST / FashionMNIST	28×28 grey images, 10 classes	days 2, 3, 7
CIFAR-10	32×32 colour images, 10 classes	day 4
MedMNIST	small medical images (X-ray, dermatology, ...)	day 5
Cora	citation graph, 2708 papers	day 8

Shoeb, PhD thesis MIT 2009 (PhysioNet) • Bhagubai et al., arXiv:2502.01224, 2025 (OpenNeuro ds005873)

The Perceptron & Representational Power

From biological to artificial neurons

- Biological neuron: dendrites collect signals, the cell "fires" if the input exceeds a threshold
- Artificial neuron: weighted sum + non-linearity
- The analogy is loose - modern networks are differentiable function approximators, not brain models



$$\hat{y} = \sigma\left(\sum_{j=1}^{d} w_j x_j + b\right) = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$$

The perceptron (Rosenblatt, 1958)

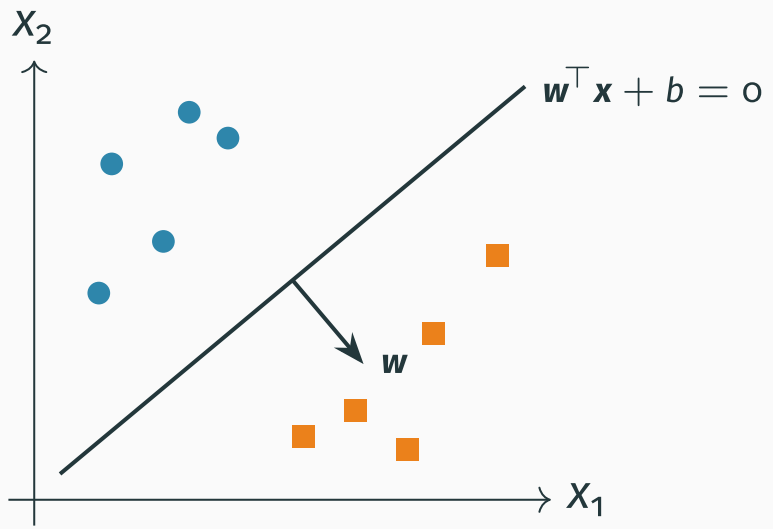
Binary classifier with a step function:

$$\hat{y} = \operatorname{sign}(\boldsymbol{w}^\top \boldsymbol{x} + b) \in \{-1, +1\}$$

Learning rule - for each misclassified sample ($y_i(\boldsymbol{w}^\top \boldsymbol{x}_i + b) \le 0$):

$$\boldsymbol{w} \leftarrow \boldsymbol{w} + \eta\, y_i \boldsymbol{x}_i, \qquad b \leftarrow b + \eta\, y_i$$

- Converges if the data are linearly separable (at most $(R/\gamma)^2$ mistakes)



Rosenblatt, Psychological Review 1958 • Novikoff 1962 (convergence proof)

Your turn: be a perceptron (15 min)

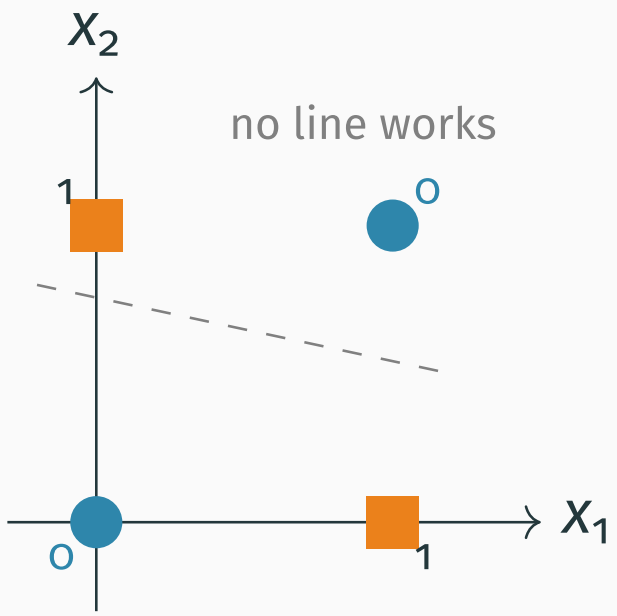
Pen and paper, in pairs.
Inputs $x_1, x_2 \in \{0, 1\}$, output $\hat{y} = 1$ if $w_1 x_1 + w_2 x_2 + b > 0$, else 0.

x_1	x_2	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

👥 Your turn

1. find w_1, w_2, b for **AND**
2. find w_1, w_2, b for **OR**
3. now try **XOR**. What goes wrong?

The limits of a single neuron: XOR



- XOR is not linearly separable ⇒ no perceptron solves it
- Minsky & Papert (1969) made this famous → interest in neural networks collapsed
- Solution: a hidden layer that transforms the input space so that the classes become separable
- But: how to train hidden layers? → backpropagation (day 2)

❓ Think about it

Which two features $h_1(\boldsymbol{x}), h_2(\boldsymbol{x})$ would make XOR linearly separable? (Hint: think of AND and OR.)

Universal approximation theorem

Theorem (Cybenko 1989; Hornik 1991)

Let σ be a non-polynomial continuous activation (e.g. sigmoid, ReLU). For every continuous $f : K \to \mathbb{R}$ on a compact set $K \subset \mathbb{R}^d$ and every $\varepsilon > 0$ there is a network with one hidden layer

$$g(\boldsymbol{x}) = \sum_{j=1}^{N} \alpha_j \, \sigma(\boldsymbol{w}_j^\top \boldsymbol{x} + b_j) \quad \text{with} \quad \sup_{\boldsymbol{x} \in K} |f(\boldsymbol{x}) - g(\boldsymbol{x})| < \varepsilon.$$

What it says

- Neural networks are expressive enough in principle

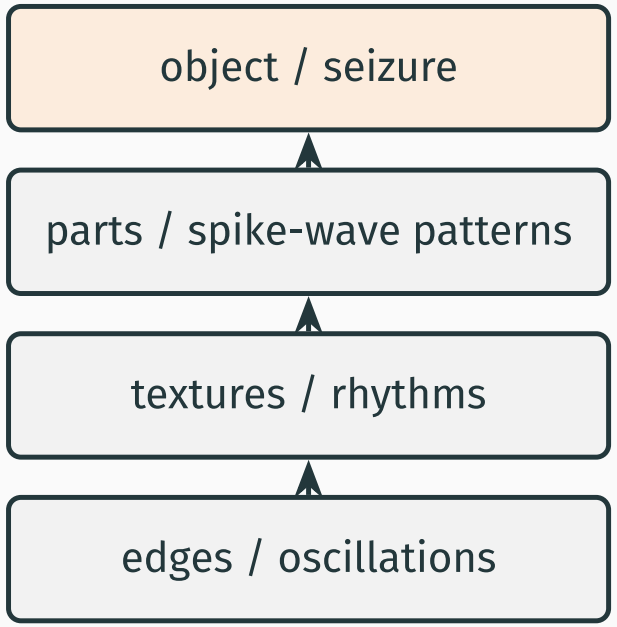
What it does not say

- how large N must be (can be exponential in d)
- that gradient descent finds g
- that g generalises to new data

Cybenko, Math. Control Signals Syst. 1989 • Hornik, Neural Networks 1991 • Leshno et al., Neural Networks 1993

Why deep and not just wide?

- ReLU networks are piecewise linear; the number of linear regions can grow exponentially with depth but only polynomially with width
- There are functions that a deep network represents compactly but a shallow one needs exponentially many units for
- Depth = composition: re-use of intermediate features



Montúfar et al., NeurIPS 2014 • Telgarsky, COLT 2016

Try it: XOR in TensorFlow Playground

playground.tensorflow.org - dataset Exclusive or

1. remove all hidden layers, inputs X_1, X_2 only. Press play. What happens?
2. add the input $X_1 X_2$. Now?
3. back to X_1, X_2 only, but one hidden layer with 4 neurons. Now?
4. dataset Spiral: what is the smallest network that solves it?
5. set the learning rate to 3 or 10. What happens to the loss?

👥 Your turn (15 min)

In your browser. Note your answers; we compare afterwards.

The three big challenges (preview)

Optimisation

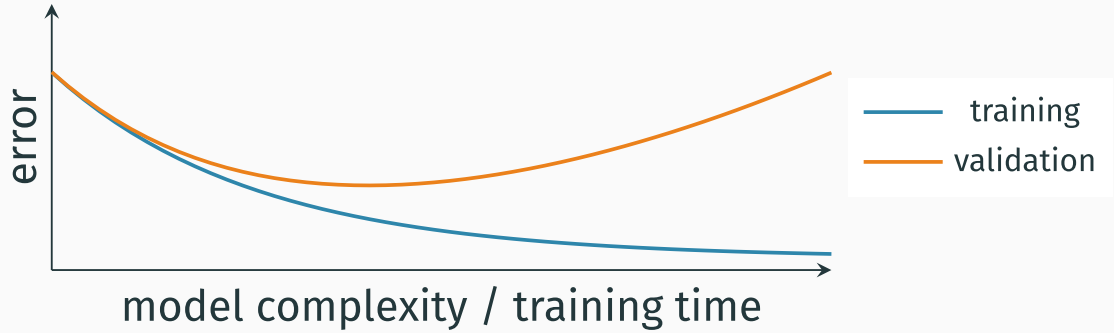
Deep networks are hard to train: vanishing / exploding gradients, bad initialisation, saddle points.
→ days 2, 3, 6

Overfitting

Millions of parameters can memorise the training set (even random labels!).
→ day 3: regularisation

Generalisation

Does it work on a new patient, a new hospital, a new device?
→ days 3, 5: evaluation, domain shift



Zhang et al., ICLR 2017 ("Understanding deep learning requires rethinking generalization")

Linear Regression as a Neural Network

Three kinds of machine learning

🏷️ Supervised

Data: pairs $(\boldsymbol{x}, y)$ - input with label

Goal: learn $f(\boldsymbol{x}) \approx y$

EEG: 4-s window → seizure yes / no

Also: spam filter, house prices, image classes

🔍 Unsupervised

Data: only $\boldsymbol{x}$ - no labels

Goal: find structure: groups (clustering), compression, generate new samples

EEG: find typical patterns or unusual segments in 1000 h of unlabelled EEG

Also: customer groups, autoencoders, GANs

🎮 Reinforcement

Data: no fixed dataset - an agent acts and gets a reward

Goal: learn the actions that bring the most reward

EEG: brain stimulation that learns when to stimulate (research)

Also: AlphaGo, robots, fine-tuning chatbots with human feedback

🔑 Key takeaway

Most of this course is supervised (today: regression). Unsupervised / generative models: day 7. Self-supervised = labels come from the data itself (e.g. predict a masked part of the EEG) - the basis of LLMs and EEG foundation models.

Regression vs. classification

Both are supervised. The difference is the type of the target y:

	Regression	**Classification**
target y	a real number, $y \in \mathbb{R}$	one class from a fixed set, $y \in \{1, \dots, K\}$
question	"how much?"	"which one?"
output of the network	the number itself (no activation)	a probability per class (sigmoid / softmax, day 2)
loss	mean squared error (MSE)	cross-entropy (day 2)
EEG example	age of the patient	seizure: yes / no
metrics	MAE, RMSE	accuracy, sensitivity, AUPRC (day 3)

🔑 Key takeaway

Same neuron $\boldsymbol{w}^\top \boldsymbol{x} + b$ - only the output and the loss change. This morning's perceptron (sign) = classification. Now: regression.

Your turn: regression or classification? (3 min)

👥 Your turn

Stand up = classification • **stay seated** = regression

1. Seizure in this 4-s EEG window: yes or no?
2. Age of the patient from the EEG
3. Sleep stage of a 30-s EEG epoch (wake, N1, N2, N3, REM)
4. Minutes until the next seizure
5. Heart rate from 10 s of ECG
6. Cat, dog or horse in the photo?

Tricky - discuss:

7. Pain score from 0 to 10
8. "Logistic regression" - which of the two is it?

Bonus - not supervised. Which kind?

9. Find groups in 1000 h of unlabelled EEG
10. A program learns a video game from its score

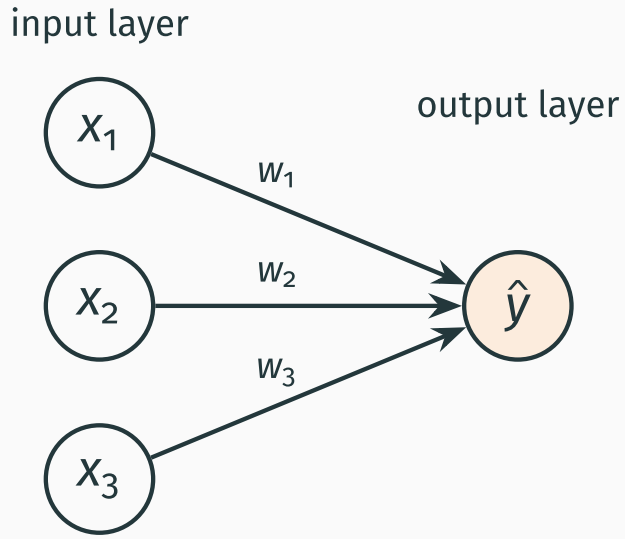
Linear regression = a single neuron without activation

Data: $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$, $\boldsymbol{x}_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$

Model:

$$\hat{y} = \boldsymbol{w}^\top \boldsymbol{x} + b = \sum_{j=1}^{d} w_j x_j + b$$

- Parameters $\boldsymbol{\theta} = (\boldsymbol{w}, b)$: $d + 1$ numbers
- Activation = identity
- Example: predict the age of a patient from EEG band powers



The loss: mean squared error

$$\mathcal{L}(\boldsymbol{w}, b) = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\boldsymbol{w}^\top \boldsymbol{x}_i + b - y_i\right)^2$$

Why squares? Assume Gaussian noise $y = \boldsymbol{w}^\top \boldsymbol{x} + b + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$:

$$-\log p(y \mid \boldsymbol{x}) = \frac{1}{2\sigma^2}(y - \boldsymbol{w}^\top \boldsymbol{x} - b)^2 + \text{const.}$$

⇒ minimising MSE = maximum likelihood

❓ Think about it

What happens to MSE if 1 % of your EEG labels contain a huge artifact value? Which loss would be more robust?

Matrix form and the closed-form solution

Stack the data: $\boldsymbol{X} \in \mathbb{R}^{n \times (d+1)}$ (with a column of ones for b), $\boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{w} \in \mathbb{R}^{d+1}$:

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{w}, \qquad \mathcal{L}(\boldsymbol{w}) = \frac{1}{n}\|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2$$

Gradient:

$$\nabla_{\boldsymbol{w}}\mathcal{L} = \frac{2}{n}\boldsymbol{X}^\top(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}) \stackrel{!}{=} 0 \quad \Rightarrow \quad \boldsymbol{w}^* = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

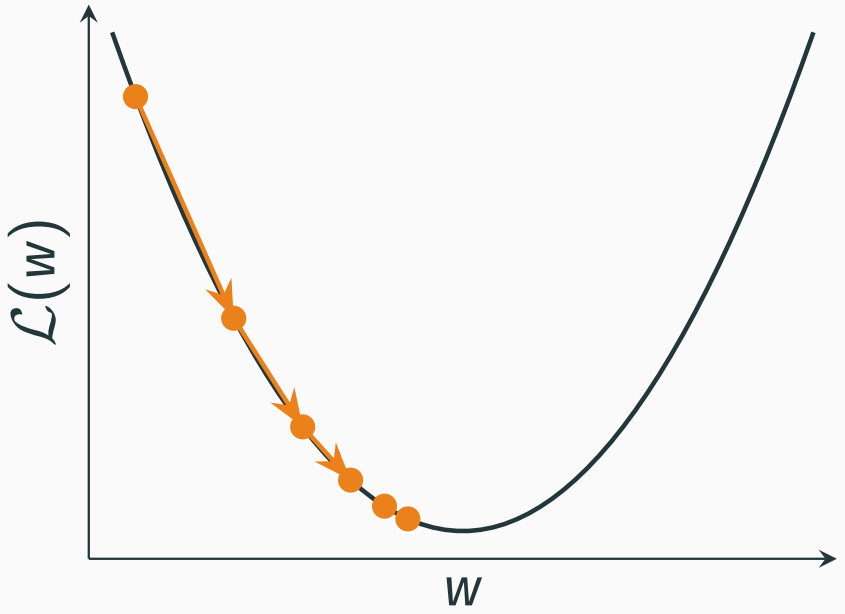
- exact, one line of code
- costs $O(nd^2 + d^3)$, all data in memory
- exists only for linear models + MSE
- deep networks: no closed form ⇒ iterative optimisation

Gradient descent

Idea: walk downhill on the loss surface.

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t)$$

- η: learning rate (step size)
- the gradient points to the steepest ascent, hence the minus sign
- stops where $\nabla \mathcal{L} = 0$ (for MSE + linear model: the global minimum, the loss is convex)



Deriving the gradient by hand

For one parameter, $\hat{y}_i = w x_i + b$, $\mathcal{L} = \frac{1}{n}\sum_i (\hat{y}_i - y_i)^2$. Chain rule:

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{1}{n}\sum_i \underbrace{2(\hat{y}_i - y_i)}_{\partial \mathcal{L}/\partial \hat{y}_i}\ \underbrace{x_i}_{\partial \hat{y}_i/\partial w}, \qquad \frac{\partial \mathcal{L}}{\partial b} = \frac{1}{n}\sum_i 2(\hat{y}_i - y_i)\cdot 1$$

- the error $(\hat{y}_i - y_i)$ is propagated back and multiplied with the input
- this is backpropagation in its simplest form (day 2: through many layers)

Always check with finite differences

$$\frac{\partial \mathcal{L}}{\partial w} \approx \frac{\mathcal{L}(w+\varepsilon) - \mathcal{L}(w-\varepsilon)}{2\varepsilon}$$

You will do this in the lab.

Your turn: one gradient-descent step by hand (15 min)

Data ($n = 3$):

x_i	0	1	2
y_i	1	3	5

Model $\hat{y} = wx + b$, start at $w = b = 0$, learning rate $\eta = 0.1$.

👥 Your turn

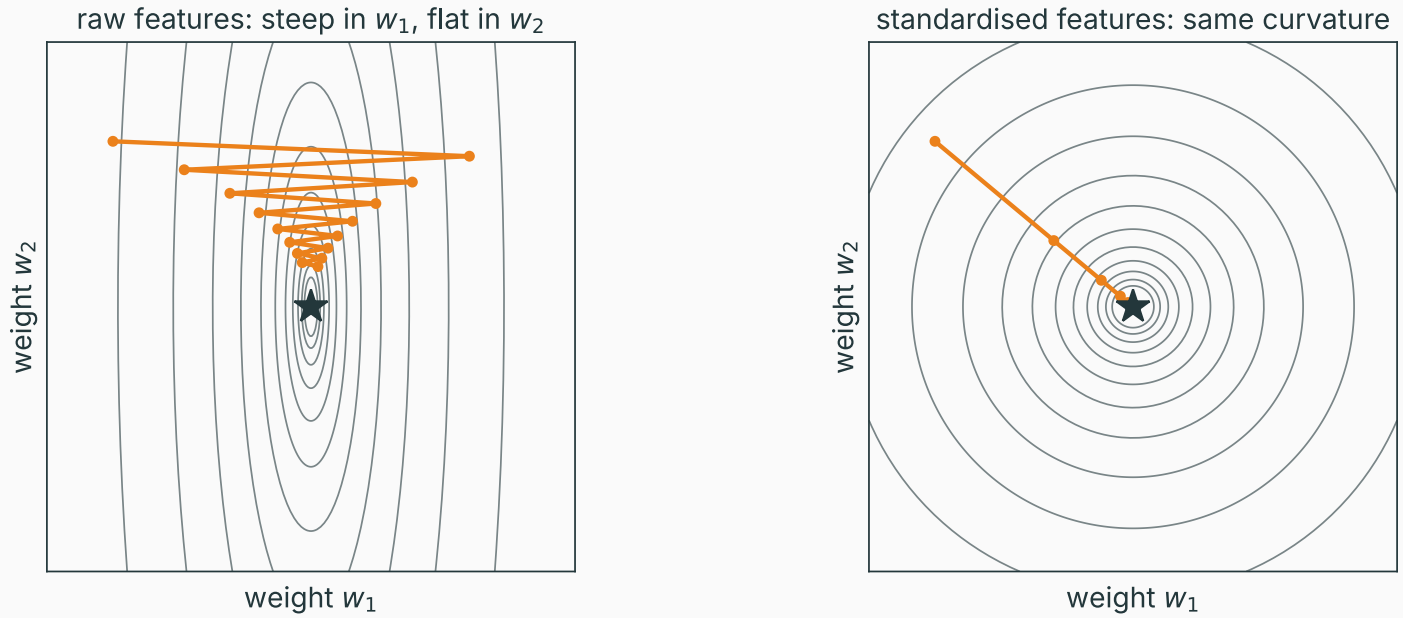
1. compute $\hat{y}_i$ and the loss $\mathcal{L}$
2. compute $\partial \mathcal{L}/\partial w$ and $\partial \mathcal{L}/\partial b$
3. update w and b
4. compute the new loss. Did it go down?

The learning rate



- too small: slow • too large: oscillates or diverges (loss becomes `nan`)
- for a quadratic with curvature c: converges iff $\eta < 2/c$ ⇒ the best η depends on the curvature

Feature scaling changes the loss landscape



- **How to read the plot:** the loss as a function of two weights, seen from above. Each line = same loss, ★ = minimum, orange = 15 gradient-descent steps
- **Example:** $\hat{y} = w_1 x_1 + w_2 x_2$ with x_1 = band power in µV² (values ≈ 1000) and x_2 = age in years (≈ 10). For MSE the curvature in direction w_j is $2\,\overline{x_j^2}$ ⇒ the loss is steep in w_1 and flat in w_2
- **Left:** η must be small enough for the steep direction ($\eta < 2/c_{\max}$) ⇒ the path zig-zags across the valley and crawls along it
- **Right:** after standardisation all directions have a similar curvature ⇒ the gradient points straight to the minimum

Standardisation in practice

Per feature j:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \quad \Rightarrow \quad \text{mean } 0, \text{ standard deviation } 1$$

1. compute μ_j, σ_j on the **training set only**
2. apply the same μ_j, σ_j to validation and test data
3. store them with the model - new data later needs them too

```
mu, sigma = X_train.mean(0), X_train.std(0)
X_train = (X_train - mu) / sigma
X_val   = (X_val   - mu) / sigma   # same mu, sigma
X_test  = (X_test  - mu) / sigma   # same mu, sigma
```

You do this in the lab (exercise 4.1).

❓ Think about it

Why not compute μ, σ on all data at once? Or separately on the test set?

Stochastic and mini-batch gradient descent

The full gradient needs all n samples: $\nabla \mathcal{L} = \frac{1}{n} \sum_{i=1}^{n} \nabla \ell_i$ - too expensive for large n.

Mini-batch SGD: estimate it on a random batch $\mathcal{B}$ of size B:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla_{\boldsymbol{\theta}} \ell_i(\boldsymbol{\theta})$$

- unbiased but noisy gradient estimate
- **iteration**: one update • **epoch**: one pass over the data (n/B iterations)
- noise can even help to escape bad regions

B	**updates/epoch**	**per update**
1	many	noisy, serial
32-512	medium	GPU-parallel
n	one	exact, slow

The algorithm

Mini-batch SGD

1. initialise $\boldsymbol{\theta}$
2. for each epoch:
   i. shuffle the data
   ii. for each batch $\mathcal{B}$:
      - forward: $\hat{\boldsymbol{y}}_{\mathcal{B}} = f_{\boldsymbol{\theta}}(\boldsymbol{X}_{\mathcal{B}})$
      - loss: $\mathcal{L}_{\mathcal{B}}$
      - backward: $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{B}}$
      - update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{B}}$
3. evaluate on validation data

```
model = nn.Linear(d, 1)
opt = torch.optim.SGD(model.parameters(), lr=0.01)
loader = DataLoader(train_ds, batch_size=64,
                    shuffle=True)
for epoch in range(epochs):
    for xb, yb in loader:
        loss = F.mse_loss(model(xb), yb)  # forward
        opt.zero_grad()                   # reset
        loss.backward()                   # backward
        opt.step()                        # update
```

This loop is the same for every model in this course.

Multiple outputs: a linear layer

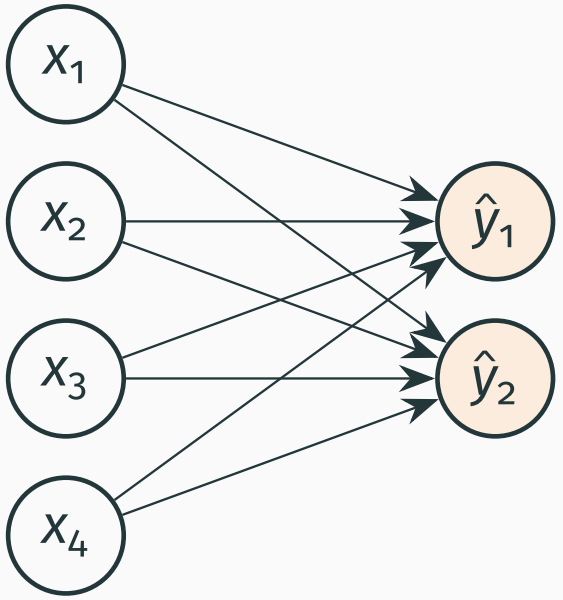
Predict k targets at once:

$$\hat{\boldsymbol{y}} = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{b}, \qquad \boldsymbol{W} \in \mathbb{R}^{k \times d},\ \boldsymbol{b} \in \mathbb{R}^{k}$$

For a mini-batch $\boldsymbol{X} \in \mathbb{R}^{B \times d}$ (one sample per row):

$$\hat{\boldsymbol{Y}} = \boldsymbol{X}\boldsymbol{W}^{\top} + \boldsymbol{b} \ \in \mathbb{R}^{B \times k}$$

- exactly `nn.Linear(d, k)`: `weight` has shape (k, d)
- one matrix product per batch ⇒ fast on GPUs



fully connected: $k \cdot d$ weights

Weight initialisation

- Linear regression with MSE is convex: any initialisation (even zeros) reaches the same minimum
- With hidden layers (day 2) this breaks down:
  - all weights equal ⇒ all hidden units compute the same thing and receive the same gradient ⇒ they stay identical forever (symmetry)
  - ⇒ initialise **randomly** to break the symmetry
- The scale matters: too large → activations explode, too small → signals vanish
- PyTorch already uses a sensible default (Kaiming-uniform for `nn.Linear`); day 3: Xavier & He initialisation

❓ Think about it

You initialise every weight of a 2-layer network with 0.5. After 1000 training steps, how many different hidden units does your network have?

Summary of the morning

🔑 Key takeaway

- Deep learning = learning representations with many differentiable layers
- One neuron = weighted sum + non-linearity; one neuron alone cannot solve XOR
- Linear regression is the simplest neural network: `nn.Linear` + MSE
- Gradient descent: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla \mathcal{L}$; mini-batches make it scale
- Learning rate, feature scaling and initialisation decide whether training works

Lab 1: PyTorch & Linear Networks

Lab 1

💻 `day1_linear_networks.ipynb`

Tasks

1. tensors, broadcasting, GPU
2. linear regression in NumPy, gradient check
3. gradient descent, learning-rate experiment
4. the same with autograd
5. `nn.Linear` with 2 outputs, batch sizes
6. first look at a seizure in CHB-MIT

Getting started

1. open the notebook in Colab
2. Runtime → Change runtime type → T4 GPU
3. work in pairs, run every cell
4. stuck > 10 min? Ask!

Autograd in five lines

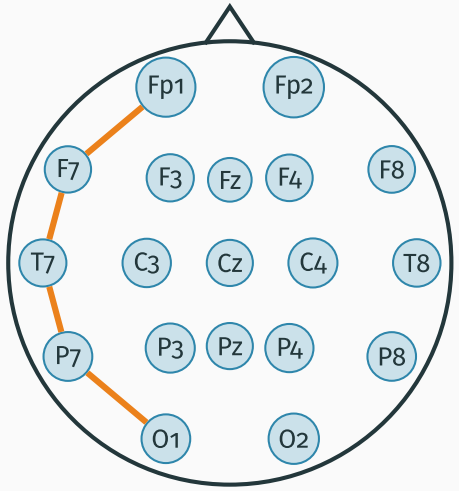
```
w = torch.tensor(0.5, requires_grad=True)   # track operations on w
b = torch.tensor(-1.0, requires_grad=True)
loss = ((w * x + b - y) ** 2).mean()        # builds a computation graph
loss.backward()                             # dL/dw, dL/db via the chain rule
print(w.grad, b.grad)                       # compare with your hand-derived gradient!
```

Two classic bugs

- forgetting `opt.zero_grad()` → gradients accumulate over steps
- updating parameters outside `torch.no_grad()` → the update becomes part of the graph

EEG in one slide

- electrodes on the scalp (10-20 system) measure voltage fluctuations (~10-100 µV)
- **bipolar montage**: differences of neighbouring electrodes, e.g. `FP1-F7`
- typical rhythms: delta (< 4 Hz), theta (4-8), alpha (8-13), beta (13-30 Hz)
- seizure: evolving rhythmic activity, often higher amplitude, spreading over channels



left temporal chain of the “double banana”

Lab 1b: from raw EEG files to a training pipeline

💻 `day1b_eeg_pipeline.ipynb` - CPU runtime is enough, about 340 MB download

1. **CHB-MIT**: files, channels, seizure statistics, spectrum (60 Hz line noise!)
2. **SeizeIT2**: BIDS, annotations, wearable EEG + ECG (heart rate during a seizure)
3. **Preprocessing**: filter, resample, windows, labelling rule, artifact checks
4. **Splits**: record-wise split, PyTorch `Dataset` / `DataLoader`, oversampling the rare class
5. **Baseline**: band power + logistic regression - sensitivity, false alarms/h, AUPRC

👥 Your turn (until the end of the lab)

Continue in pairs after Lab 1. What you do not finish today is homework before day 2.

🧠 Why this matters

This is the template for your project: data, preprocessing, split, baseline, metrics.

Common pitfalls (learn from others' mistakes)

- data leakage: windows of the same patient in train and test
- tuning on the test set
- accuracy on imbalanced data
- comparing to numbers from a different split
- starting to write the paper in the last week
- underestimating data preprocessing (it takes the longest!)
- no fixed seeds / versions ⇒ results cannot be reproduced
- one person does all the code

🧠 EEG corner

Random window splits on CHB-MIT can give near-perfect scores that collapse on a new patient. We will measure this effect ourselves on day 3.

Wrap-up

Today

🔑 Key takeaway

- DL learns representations; the ideas are old, data + GPUs + software made them work
- Perceptron → XOR problem → hidden layers; UAT: expressive, but no free lunch
- Linear regression = one linear layer + MSE, trained by (mini-batch) gradient descent
- The PyTorch training loop: forward → loss → `zero_grad` → backward → step

Self-study & project milestone

Read / watch

- D2L, chapters 2 (preliminaries) and 3 (linear neural networks for regression)
- Nielsen, chapter 1
- 3Blue1Brown, Neural networks, videos 1-2 (3blue1brown.com)
- Seeing Theory, chapter Regression Analysis (seeing-theory.brown.edu)
- finish the lab notebook
- all links: course repository (QR code)

👥 Exam

- teams of two
- the challenge starts on day 4

Questions?

Backup

Extra material - if we are faster than planned

Backup: what fits where

Slide	Time	Fits best
Quick quiz	10-15 min	end of the morning, or start of the afternoon
Perceptron learning rule by hand	10-15 min	after "The perceptron"
MSE = maximum likelihood	10 min	after "The loss: mean squared error"
Normal equations by hand	10-15 min	after "Matrix form" or after the GD-by-hand exercise
Beyond straight lines: under- and overfitting	10-15 min	after "The three big challenges"
Robust losses	10 min	after "The loss: mean squared error"
Discussion: seizure detection in practice	15 min	after "Running example" or before the project part
Dissect a paper in 15 minutes	20-30 min	project kick-off
Bonus lab challenges	open	during the lab, for fast pairs

Backup: quick quiz

Show of hands or shout it out - one minute per question.

1. How many parameters does `nn.Linear(7, 2)` have?
2. Your loss becomes `nan` after three steps. What do you try first?
3. Why do we compute mean and std for standardisation on the training set only?
4. Can a single perceptron learn OR? XOR?
5. You forget `opt.zero_grad()`. What happens?
6. 3200 training samples, batch size 32, 10 epochs: how many parameter updates?
7. A seizure detector reaches 98 % accuracy on CHB-MIT windows. Good?

Backup: perceptron learning rule by hand

	x_1	x_2	y
A	1	2	+1
B	2	1	−1
C	0	1	+1
D	2	0	−1

Start at $\boldsymbol{w} = (0, 0)$, $b = 0$, $\eta = 1$. Visit A, B, C, D in this order, repeat until no mistake.

👥 Your turn (10 min, pen and paper)

For each point: compute $a = \boldsymbol{w}^\top \boldsymbol{x} + b$. If $y \cdot a \leq 0$ update $\boldsymbol{w} \leftarrow \boldsymbol{w} + y\boldsymbol{x}$, $b \leftarrow b + y$.

1. How many updates until all points are correct?
2. Draw the final decision line $\boldsymbol{w}^\top \boldsymbol{x} + b = 0$.
3. What would happen with the XOR points?

Backup: MSE = maximum likelihood, step by step

Assume $y_i = \boldsymbol{w}^\top \boldsymbol{x}_i + b + \epsilon_i$ with independent $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$:

$$p(y_i \mid \boldsymbol{x}_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(- \frac{(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b)^2}{2\sigma^2} \right)$$

Likelihood of the whole dataset = product; take $-\log$:

$$-\log \prod_{i=1}^{n} p(y_i \mid \boldsymbol{x}_i) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b)^2$$

The first term and $\frac{1}{2\sigma^2}$ do not depend on $\boldsymbol{w}, b$ ⇒ maximising the likelihood = minimising the sum of squared errors.

Backup: normal equations by hand

$$\mathcal{L}(\boldsymbol{w}) = \tfrac{1}{n}(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})^\top(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}) = \tfrac{1}{n}\left(\boldsymbol{w}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{y}^\top\boldsymbol{X}\boldsymbol{w} + \boldsymbol{y}^\top\boldsymbol{y}\right)$$

$$\nabla_{\boldsymbol{w}}\mathcal{L} = \tfrac{2}{n}\left(\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{w} - \boldsymbol{X}^\top\boldsymbol{y}\right) = 0 \quad \Rightarrow \quad \boldsymbol{X}^\top\boldsymbol{X}\,\boldsymbol{w} = \boldsymbol{X}^\top\boldsymbol{y}$$

Same data as the GD exercise:

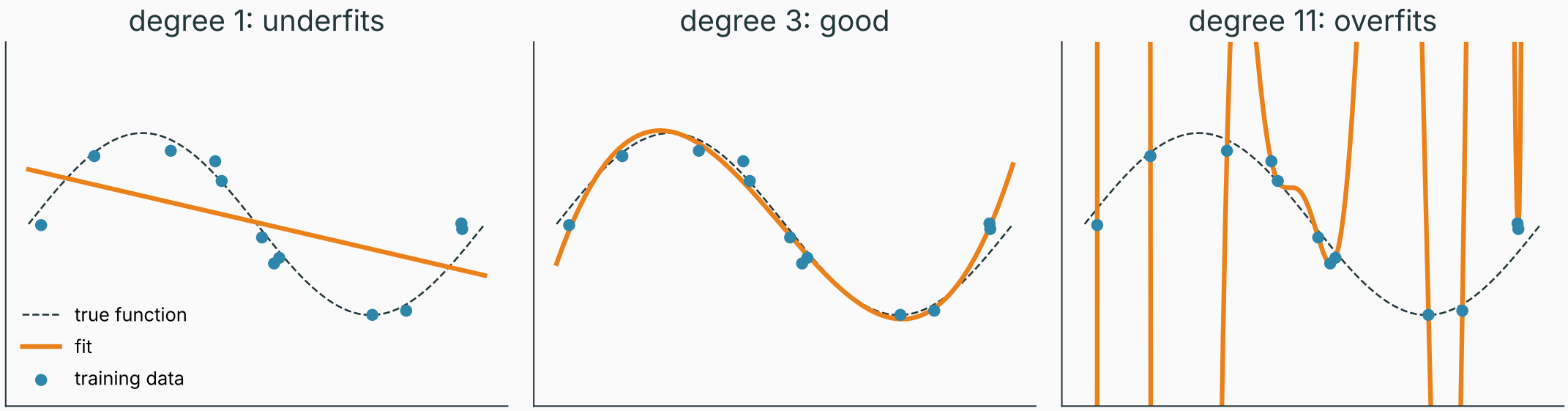
x_i	0	1	2
y_i	1	3	5

👥 Your turn (10 min)

1. write down $\boldsymbol{X}$ (with a column of ones) and $\boldsymbol{y}$
2. compute $\boldsymbol{X}^\top\boldsymbol{X}$ and $\boldsymbol{X}^\top\boldsymbol{y}$
3. solve the 2×2 system. Compare with your gradient-descent step.

Backup: beyond straight lines - under- and overfitting

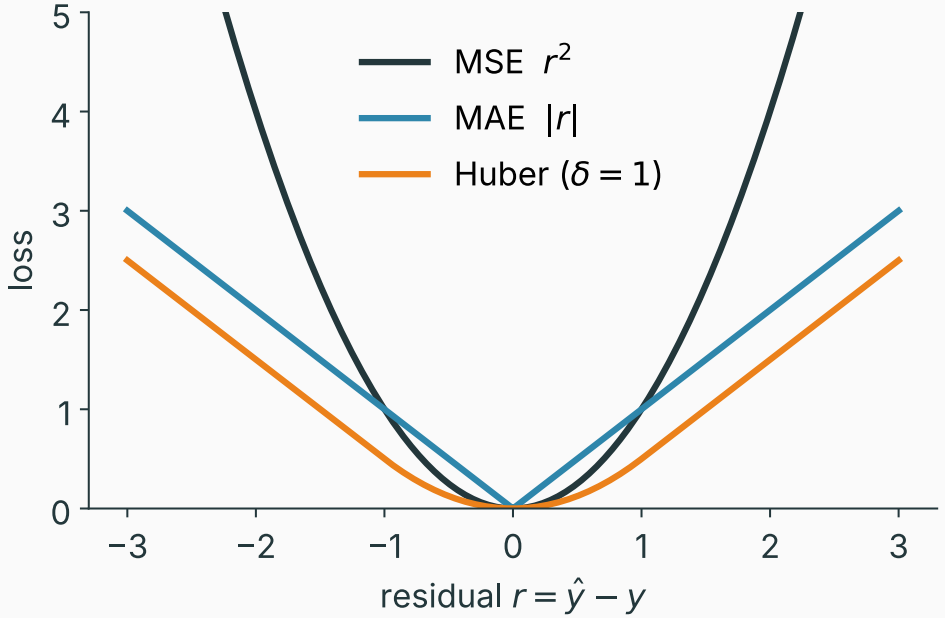
A linear model in features $\phi(x) = (x, x^2, \ldots, x^p)$ is still linear in $\boldsymbol{w}$ - same loss, same gradient descent.

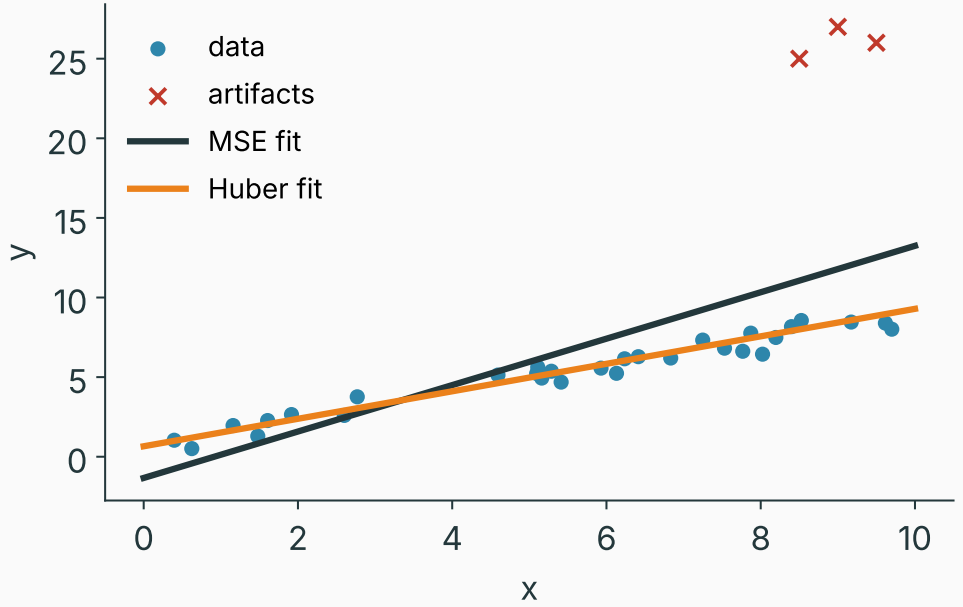


👥 Try it (5 min)

TensorFlow Playground, dataset Spiral: noise 50, ratio of training data 10 %, a large network, check Show test data. Watch training vs. test loss.

Backup: robust losses





- MSE grows quadratically: three artifact labels pull the whole line
- MAE / Huber grow linearly for large residuals ⇒ outliers have bounded influence
- In PyTorch: `nn.MSELoss`, `nn.L1Loss`, `nn.HuberLoss(delta=1.0)`

Backup: discussion - seizure detection in practice

A wearable seizure detector can be tuned to one of two operating points:

- **A:** detects 90 % of seizures, 1 false alarm per hour
- **B:** detects 70 % of seizures, 1 false alarm per day

1. Which one for a patient who lives alone?
2. Which one for an automatic seizure diary in a drug trial?
3. What does a missed seizure cost, what does a false alarm cost? (For the patient, for the doctor, for the trial.)
4. Who should choose the threshold - the developer, the doctor, the patient?

👥 In groups of 3-4 (10 min)

Agree on an answer to each question.

Backup: dissect a paper in 15 minutes

Pick any deep-learning paper and answer:

1. **Task:** what goes in, what comes out?
2. **Data:** which dataset, how is it split?
3. **Model** in one sentence
4. **Baseline:** what do they compare against?
5. **Metric** and main result (one number)
6. **Evidence:** is the main claim supported by a table or figure?
7. **Practical:** code available? Fits on a Colab T4? Can the idea be added to a simple baseline?

👥 In your (future) groups (15 + 10 min)

15 min reading - then each group presents its paper in 1 minute.

🧠 Tip

Read in this order: abstract → figures and tables → conclusion → method.

Backup: bonus lab challenges

For pairs who finish early - all in today's notebook, no extra data needed.

1. **Adam vs. SGD:** call `train_linear(optimiser="adam")` and compare the test MSE curves. Which learning rate does Adam need?
2. **Outliers:** comment out the outlier filter in Part 4 and re-run batch size 1. What happens, and why?
3. **Robust loss:** replace `mse_loss` with `l1_loss` or `huber_loss` in `train_linear`. Compare on data with and without outliers.
4. **Polynomial features:** fit the synthetic data of Part 2 with features $x, x^2, \ldots, x^9$ using only 10 points. Plot the fit.
5. **EEG spectrum:** take channel `F7-T7`, one 4-s window before and one during the seizure. Plot both power spectra (`np.fft.rfft`). Which frequency band changes?